Plate-fin heat sink

Request: a heat sink for a 40 W chip in still air

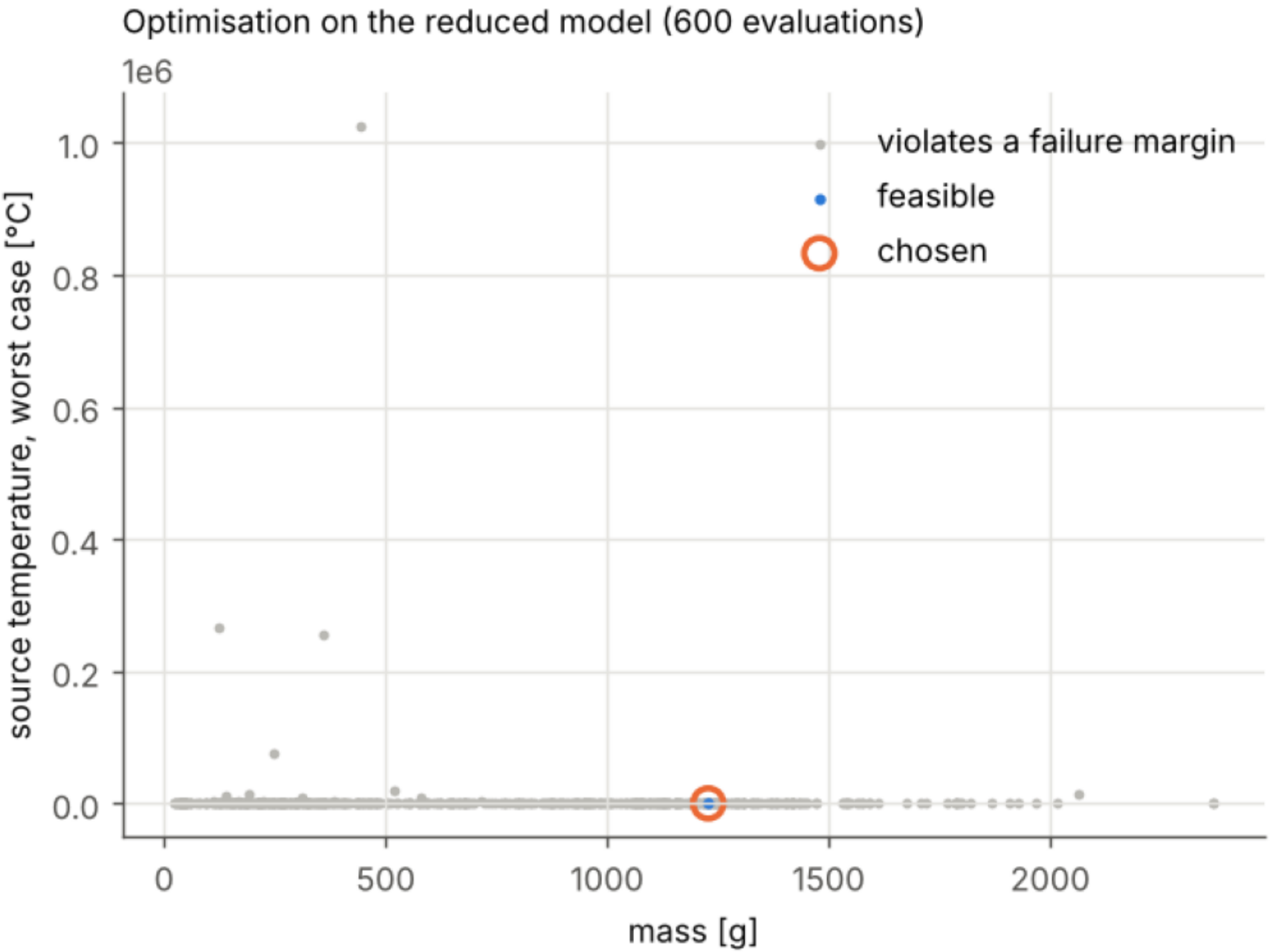
Verdict: will work   P = 1.000 (95 % CI 0.998-1.000)   trust: APPROVED

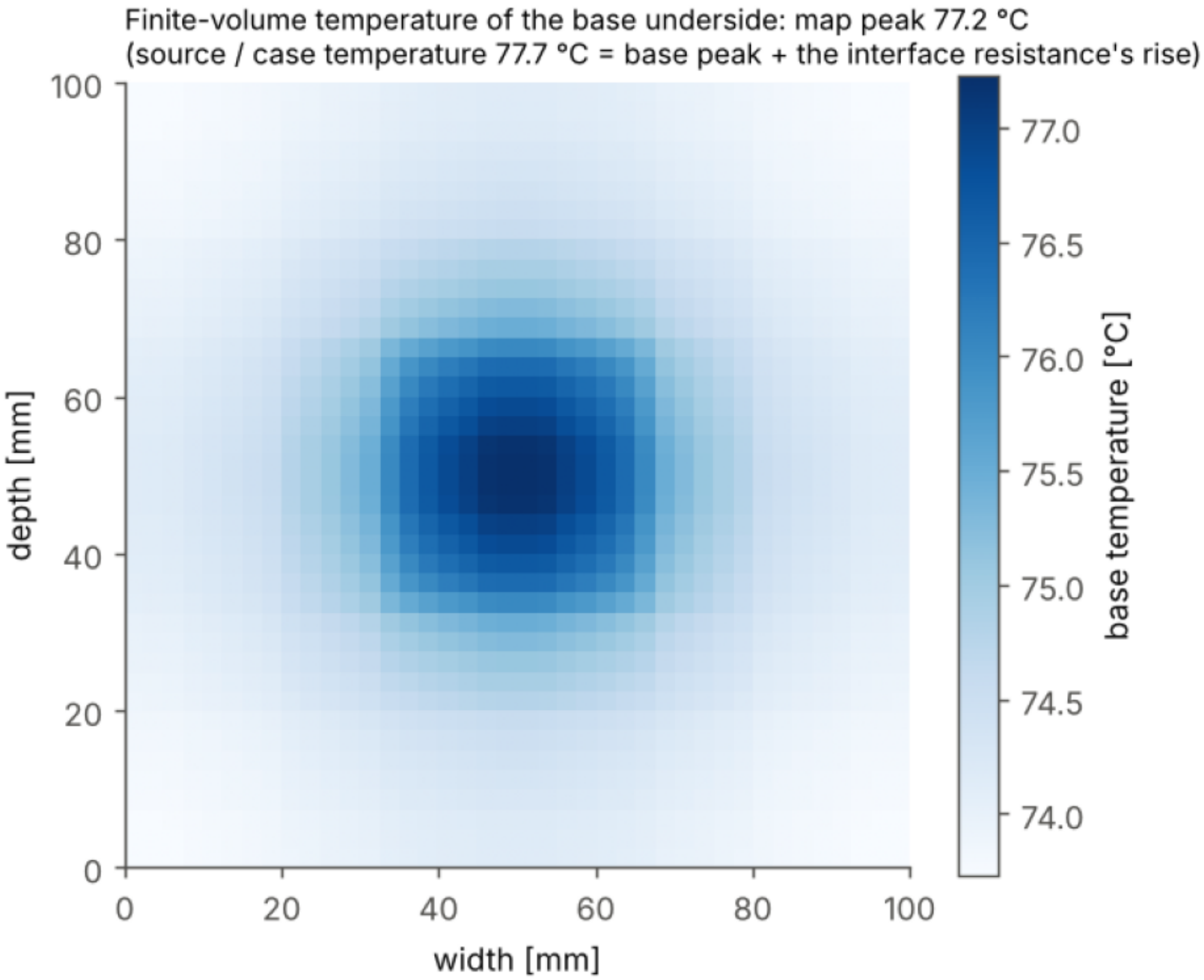
Design:
   base width: 100 mm
   base depth (along the fins): 100 mm
   base thickness: 6.5 mm
   number of fins: 14
   fin thickness: 1.1 mm
   fin height: 47 mm
   material: cu110

Numbers (fidelity):
   fins can be extruded (gap): 3.338   [resistance network]
   source stays below its limit: 0.0171   [resistance network]
   r_total_k_w: 1.475 K/W [resistance network]
   fins can be extruded (aspect): 1.769   [resistance network]
   h_w_m2k: 5.161 W/(m²·K) [resistance network]
   fin_gap_mm: 6.508 mm [resistance network]
   t_source_c: 83.99 °C [resistance network]
   fin_efficiency: 0.9822 - [resistance network]
   mass_g: 1228 g [resistance network]
   source stays below its limit: 0.04192   [hand calculation]
   fins can be extruded (gap): 3.338   [hand calculation]
   fins can be extruded (aspect): 1.769   [hand calculation]
   t_source_c: 77.68 °C [3-D finite volume (base)]
   t_source_band_hi_c: 83.85   [3-D finite volume (base)]
   r_total_k_w: 1.317 K/W [3-D finite volume (base)]
   margin_temp: 0.139   [3-D finite volume (base)]
   mass_g: 1228 g [3-D finite volume (base)]

Verification:
   t_source_c: full 77.68, reduced 77.59, rel. err 0.00108

Limits:
   - Correlation-based convection: +-15 % on h is the stated band of the correlations, not a measured uncertainty of
     this part.
   - Radiation is neglected (conservative).
   - The full model resolves conduction in the base; the fins and the air are modelled by the correlations, not by
     CFD.
   - The source temperature is the case / lid temperature; the junction is hotter by the package's own resistance (not
     modelled).


Optimisation on the reduced model (600 evaluations)
1e6
1.0
0.8
0.6
0.4
0.2
0.0
source temperature, worst case [°C]
violates a failure margin
feasible
chosen
0
500
1000
1500
2000
mass [g]


Finite-volume temperature of the base underside: map peak 77.2 °C
(source / case temperature 77.7 °C = base peak + the interface resistance's rise)
depth [mm]
width [mm]
base temperature [°C]
0
20
40
60
80
100
74.0
74.5
75.0
75.5
76.0
76.5
77.0

